MIL-W-8957A
13 May 1966
Superseding
MIL-W-8957(ASG)
15 January 1965

MILITARY SPECIFICATION

WIRE, STEEL, CARBON, HIGH STRENGTH

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers a premium quality round wire of highcarbon steel.

1.2 Composition.- The chemical composition of the wire shall conform to table I (see 3.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

UU-P-271	Paper, Wrapping, Waterproofed Kraft

Military

MIL-B-121	Barrier Material, Greaseproofed, Waterproofed, Flexible

STANDARDS

Federal

Fed. Test Method Std. No. 151	Metals, Test Methods

Military

MIL-STD-163	Steel Mill Products Preparation for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 9505

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2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

American Society for Testing and Materials

ASTM A318-56 Tension Testing of Steel Spring Wire

(Copies of ASTM publications may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

Naval Air Engineering Center

Drawing D-11711 Spool, Towline, Shipping, 500 Pounds Capacity

(Copies of NAEC drawings may be obtained from the Aeronautical Materials Laboratory, Naval Air Engineering Center, Philadelphia, Pennsylvania 19112.)

3. REQUIREMENTS

3.1 Chemical composition.- The composition shall conform to the respective limits of table I. Individual determinations may vary from the specified range to the extent indicated in the check analysis column, except that elements in any heat shall not vary both above and below the specified range.

TABLE I. Chemical composition

Element	Range (percent)	Check analysis tolerance (percent)
Carbon	0.85 - 0.90	±0.02
Manganese	.25 - 0.40	±0.02
Silicon	.15 - 0.25	±0.02
Phosphorous	.025 max.	+0.005
Sulfur	.020 max.	+0.005
Chromium	.04 max.	—
Molybdenum	.04 max.	—
Copper	.08 max.	—
Nickel	.03 max.	—
Nitrogen	—	—
Carbon-nitrogen	—	—

3.2 Dimensions - diametral tolerances.- The permissible variation from the nominal diameter shall be ±3 percent of the diameter. The tolerance on out-of-roundness shall be half the diametral tolerance.

3.3 Mechanical properties - torsional, ductility, and tensile.- The torsion, ductility, and tensile breaking strength minimum requirements shall be as specified in table II for various wire diameters, when tested as specified in 4.3.3.2, 4.3.3.3, 4.3.4.2, and 4.3.5.2, respectively.

3.4 Joints.- Wire rods and wires may be joined by welding, provided such operations are accomplished in the initial stages of processing, and that variations in microstructure due to welding are eliminated by repeated annealing and drawing operations.

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TABLE II. Ductility, torsion, and tensile strength requirements

Wire diameter (inch)	Torsion, number of 360° twists	Ductility test		Tensile breaking strength 1/	
		Sheave diameter (inch) 2/	Sheave center spacing (inch) 2/	Stress (ksi)	Load (pounds)
0.0032 -0.0035	---	---	---	600	---
.004	---	---	---	575	7.0
.005	---	---	---	545	10.8
.006	---	---	---	540	15
.007	---	---	---	530	---
.008	---	---	---	525	26
.009	---	---	---	505	---
.010	90	---	---	495	39
.011	---	---	---	480	---
.012	85	---	---	475	53
.013	---	---	---	460	---
.014	75	---	---	450	69
.015	---	---	---	440	---
.016	75	---	---	440	90
.017	---	---	---	435	---
.018	60	---	---	425	102
.019	---	---	---	420	---
.020	60	---	---	415	125
.021	---	---	---	410	---
.022	60	---	---	405	152
.023	---	---	---	400	---
.024	60	---	---	395	180
.025	50	0.866	1.0	395	190
.026	---	---	---	393	---
.027	---	---	---	391	---
.028	50	0.866	1.0	390	240
.029	---	---	---	388	---
.030	---	---	---	385	---
.031	---	---	---	382	---
.032 - .033	48	0.867	1.0	380	302
.034 - .035	---	---	---	375	---
.036 - .037	44	1.181	1.38	372	376
.038 - .039	---	---	---	370	---
.040 - .041	44	1.181	1.38	367	465
.042 - .043	41	1.181	1.38	364	505
.044 - .045	41	1.181	1.38	360	587
.046 - .047	---	---	---	355	---
.048 - .049	37	1.181	1.38	353	624
.050 - .051	37	1.181	1.38	350	690
.052 - .053	---	---	---	347	---
.054 - .055	34	1.181	1.38	345	778
.056 - .057	30	1.457	1.62	342	824
.058 - .059	30	1.457	1.62	340	885
.060 - .061	25	1.457	1.62	335	997
.062 - .063	---	---	---	331	---
.064 - .065	25	1.457	1.62	329	1,045
.066 - .067	---	---	---	327	---
.068 - .069	---	---	---	325	---
.070 - .071	---	---	---	323	---
.072 - .073	25	1.457	1.62	321	1,322

1/ The maximum breaking load shall be not greater than 25 ksi above the permissible minimum strength.

2/ Tolerance on roll groove diameter +0.050, -0.00 inch. Tolerance on distance between centers, ±0.12 inch.

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3.5 Workmanship.- Wire shall be free from defects such as seams, pits, e marks, scrapes, cold shuts and other material or surface imperfections ıding to impair the serviceability of the wire. Wire surfaces shall be bright ł smooth.

QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the ıtract or purchase order, the supplier is responsible for the performance of l inspection requirements as specified herein. Except as otherwise specified, e supplier may utilize his own facilities or any commercial laboratory acceptable the Government. The Government reserves the right to perform any of the ;pections set forth in the specification where such inspections are deemed :essary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections.- The examining and testing of the wire ıll be classified as quality conformance inspections.

4.3 Inspection methods.-

4.3.1 Dimensional visual inspection.- Two determinations of wire diameter shall made on the wire on each shipping reel. The exposed surface of wire shall be ;ually examined for compliance with workmanship requirements while being wound the shipping reels.

4.3.2 Chemical composition

4.3.2.1 Sampling.- One sample for testing shall be selected to represent ten ipping reels of wire of one diameter and composition, except that when wire supplied ı be positively identified with the specific heat of steel from which the wire :erial was obtained, certified copies of the ladle analysis supplied by the steel ›ducer will be acceptable.

4.3.2.2. Method.- Tests shall be conducted in accordance with Method 111 or ? of Fed. Test Method Std. No. 151.

4.3.3 Torsion test.-

4.3.3.1 Sampling.- One specimen shall be selected from the first, last, and e intermediate shipping reel from each manufacturing coil at the time the wire is ing transferred to the shipping reels.

4.3.3.2 Method (wire over 0.032-inch diameter).- Specimens for the torsion ;t of wire over 0.032-inch diameter shall be not less than 10 inches long. One d of the specimen shall be clamped in the fixed head of the testing machine and e other end clamped in the movable head. The distance between the clamps in ich the wire is held shall be not less than 8 inches. The wire shall be twisted a uniform rate, in one direction, until failure occurs. The speed of the machine ıll be such as not to cause undue heating of the wire, and in no case shall the ?ed exceed 33 rpm for wire diameters 0.125 inch or less, or 20 rpm for wires of ·ger diameters. Sufficient tension shall be applied longitudinally to the wire keep it from kinking during the test.

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4.3.3.3 Method (wire 0.032-inch diameter and less).- The torsion specimen for this wire shall be not more than 30 inches long. The center of this length of wire shall be passed around a hook held in the movable head of the testing machine. The loose ends shall then be clamped together in the stationary head and the test conducted as specified in 4.3.3.2.

4.3.4 Ductility test.-

4.3.4.1 Sampling.- All the wire supplied in accordance with this specification shall be tested.

4.3.4.2 Method.- Wire shall be passed through a series of five rolls or sheaves as it is being transferred from manufacturing coils or reels to shipping reels. The sheaves shall be of the size and spacing between centers of rotation indicated by table II for the respective wire size. The sheaves shall be arranged as indicated on figure 1. Breakage of the wire shall be cause for rejection of all lengths of wire less than the minimum individual lengths specified in table III for the respective wire size. No joining or splicing or retest of shorter lengths will be permitted.

4.3.5 Tensile test.-

4.3.5.1 Sampling.- One specimen shall be selected from the first, last, and one intermediate shipping reel from each manufacturing coil at the time the wire is being transferred to the shipping reels and subsequent to the ductility test of 4.3.4.

4.3.5.2 Method.- Wire specimens shall be tested in accordance with ASTM Method A318-56.

4.3.6 Rejection criteria.- Failure of any specimen or sample to meet the inspection and test requirements specified herein shall be cause for rejection of the lot (coil) represented.

5. PREPARATION FOR DELIVERY

5.1 Preservation.- A light grade of preservative oil shall be applied to the wire surfaces. Prior to winding on the reel, a layer of waterproof material conforming to UU-P-271 and a layer of greaseproof material conforming to MIL-B-121 shall be applied to the barrel of the reel and inner surfaces of the flanges. After the wire is in place a layer each of greaseproof and waterproof barrier material shall be applied in a manner to enclose the wire, with the waterproof material on the outside. The barrier materials shall be held in place by wire bands or metal straps.

5.2 Level A, B, and C, packaging and packing.- Unless otherwise specified, wire shall be packaged, preserved, and packed for domestic or overseas shipment in accordance with MIL-STD-163, except that unit and intermediate package identification marking shall include the items indicated in 5.3. The level of packaging and packing desired shall be as specified by the procuring activity (see 6.2).

5.2.1 Level A, packaging and packing (applicable to .045 and .051 wire only).- Wire shall be of lengths as indicated in table III and shall be wound on shipping reels of dimensions as indicated on Drawing NAEC D-11711 for Navy procurement. For Air Force procurement, the wire shall be wound on reels as specified or agreed on by the procuring activity. The wire shall be evenly wound under a uniform tension of approximately 40 pounds. When winding is complete, the end shall be fastened to the flange of the reel by means of a "U" or "J" bolt.

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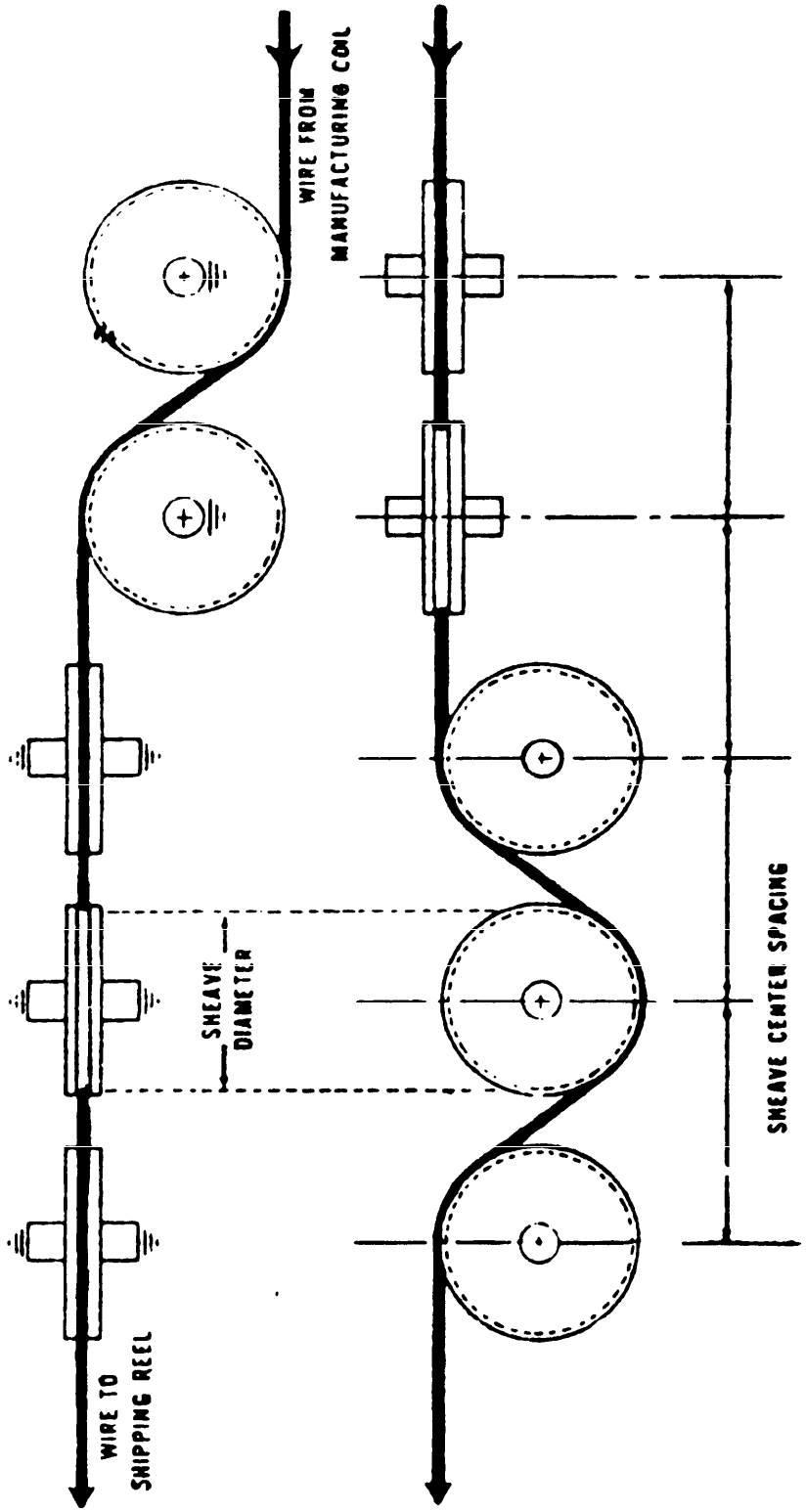


FIGURE 1. Ductility test setup

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TABLE III. Shipping reel amounts of wire

Wire diameter (inch)	Individual lengths, (feet) approximate	Total per reel, (feet) approximate
0.045	28,000, 56,000, 84,000	84,000
.051	22,000, 44,000, 66,000	66,000
.051	35,000, 70,000	70,000

5.3 Marking.- Shipping reels shall be identified with aluminum marking plates securely attached to the ends of the reels and marked with the following information:

Purchase order or contract number
Manufacturer's name
MIL-W-8957A
Wire diameter and approximate length
Lengths individual wires

In addition, a tag with the above marking information shall be firmly fixed to the outside of the waterproof wrapping paper in such manner that the information will be fully visible on reels stacked end-on-end.

6. NOTES

6.1 Intended use.- The wire is intended for use in aerial target towline and other applications requiring high-strength steel wire.

6.2 Ordering data.- Procurement documents should specify:

(a) Title, number, and date of this specification.
(b) Wire diameter, lengths of wire, and total amount of wire desired (see 1.2, tables I, II, and III).
(c) For Air Force procurement, the type of shipping reel (see 5.2.1).
(d) Applicable levels of packaging and packing (see 5.2).

Custodians:
ARMY- AR
Air Force - 20

Preparing activity:
ARMY - AR

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-W-8957A	WIRE, STEEL, CARBON, HIGH STRENGTH

NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	7b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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